DESIGN PROJECT

LLM Hallucination Detection using Multi-Layer EigenScore

PRESENTED BY

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1. Problem Statement

Large Language Models often **hallucinate**: they give confident answers that are wrong, in the same tone as correct ones.

> **Goal:** for every answer, output a **hallucination score** that says how likely it is to be wrong, **without knowing the correct answer**, and running on an **8 GB laptop GPU**.

2. Key Idea: Ask Again, Check Agreement

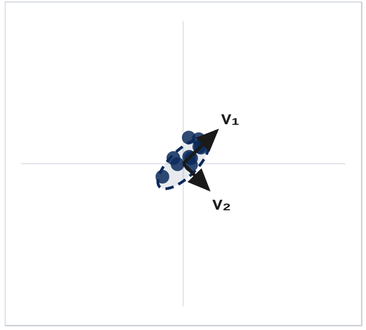
Q: "In which year did the bridge open?", asked 5 times

Model knows	1932	1932	1932	1932	1932	✓ trust
Model guesses	1932	1928	1936	1941	1930	⚠ flag

Fig. 1. Answers that agree signal knowledge; answers that scatter signal guessing.

We measure agreement not on the words but on the model's **internal representation** of each answer: 4096 numbers read from its hidden layers.

3. Measuring Spread with Eigenvalues



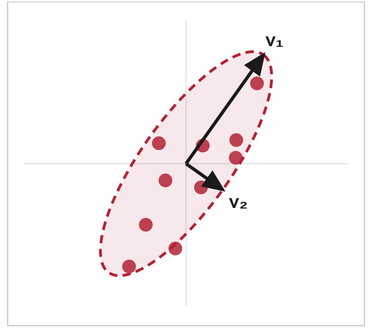
Answers agree

small cloud → small eigenvalues

Eigenvalues (spread along v_1, v_2)

λ1 = 0.04

λ2 = 0.01



Answers disagree

large cloud → large eigenvalues

Eigenvalues (spread along v_1, v_2)

λ1 = 0.66

λ2 = 0.07

Fig. 2. Each dot is one answer's internal representation. Arrows are the eigenvectors (main directions of spread); bar lengths are the eigenvalues (amount of spread).

EigenScore = average of log(eigenvalues)

Small eigenvalues → answers agree → **low score** · Large eigenvalues → answers disagree → **high score**

4. Worked Example

Toy case: **3 answers**, each represented by **2 numbers** (real system: 10 answers × 4096 numbers).

Case A: answers agree

Answer	x	y
1	2.0	1.0
2	2.1	1.1
3	1.9	0.9

Covariance = [0.01 0.01; 0.01 0.01]
Eigenvalues = 0.021, 0.001
Score = ½(log 0.021 + log 0.001)
= **−2.34 (low → trust)**

Case B: answers disagree

Answer	x	y
1	2.0	1.0
2	0.5	2.5
3	3.0	2.0

Covariance = [1.58 −0.42; −0.42 0.58]
Eigenvalues = 1.735, 0.434
Score = ½(log 1.735 + log 0.434)
= **−0.06 (high → flag)**

log = $\log_{10}$; a small constant α = 0.001 is added to the covariance so the log is always defined.

5. Method

Question to the LLM ▶ **10 answers** sampled ▶ **Hidden states** middle layers ▶ **Eigenvalues** of covariance ▶ **Score** high = flag

Fig. 3. Detection pipeline.

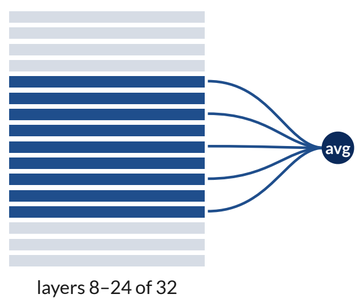


Fig. 4. Multi-layer: average the score over the middle layers instead of one layer.

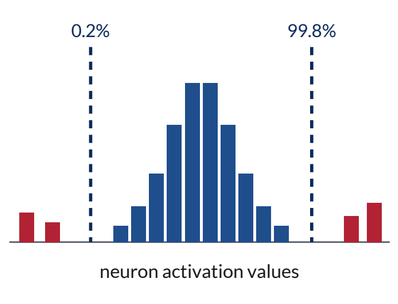


Fig. 5. Feature clipping: extreme values (red) are capped at the dashed limits.

6. Implementation

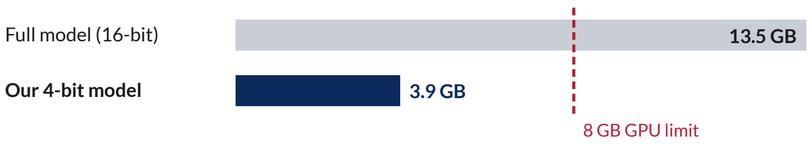


Fig. 6. 4-bit weights let a 7-billion-parameter model fit a laptop GPU.

Python, PyTorch and HuggingFace · Llama-2-7B · hooks capture only the needed layers · **8 / 8** automated tests pass.

7. Experiment

Questions tested	**250 (CoQA reading comprehension)**
Answers sampled per question	**10**
Answer counted correct if	**matches reference (ROUGE-L > 0.5)**
Hardware	**RTX 4060 laptop GPU, 8 GB**

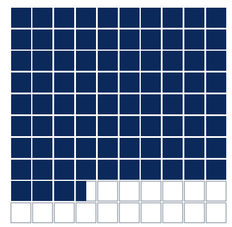
8. Results

The LLM's own answers (250 questions)

92 correct (37%) | 158 wrong (63%): need to be caught

Our system must tell these two groups apart without knowing which is which.

83.5% Take one **wrong** answer and one **correct** answer at random. In **83.5% of cases**, our system gives the wrong one the higher hallucination score.



83.5 pairs ranked correctly

16.5 pairs ranked wrongly

Random guessing would score 50.
A perfect detector would score 100.
(This measure is called AUROC.)

Fig. 7. Out of every 100 (wrong, correct) answer pairs.

Result	Value
Detection score (AUROC)	**83.5%**
With feature clipping on	**82.5%**
Time per question	**≈ 6 seconds**
GPU memory used by model	**≈ 3.9 GB**

9. Conclusion & Future Work

Our system flags likely-wrong LLM answers using only the model's internal states: **83.5% detection score**, no answer key, on an 8 GB laptop GPU.

Next: full multi-layer evaluation on 3 datasets (1,000 questions each).

10. References

[1] C. Chen et al., "INSIDE: LLMs' Internal States Retain the Power of Hallucination Detection," ICLR 2024.

[2] S. Reddy et al., "CoQA: A Conversational Question Answering Challenge," TACL 2019. [3] H. Touvron et al., "Llama 2," arXiv:2307.09288, 2023.